

CIE xyZ color space

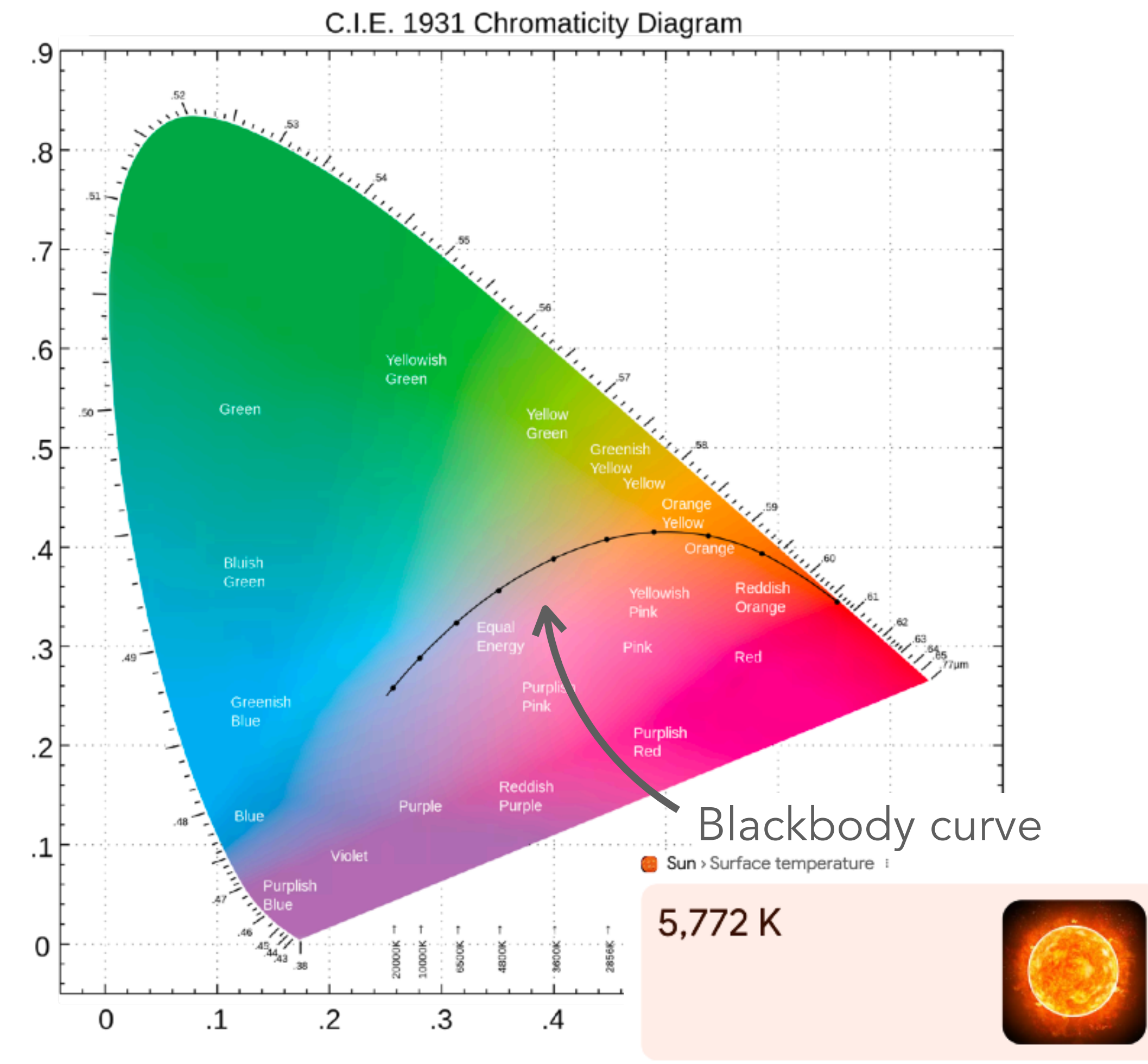
xyZ: a **chromaticity** model that isolates color (chromaticity) from luminance

$$x = \frac{X}{X + Y + Z}$$

$$y = \frac{Y}{X + Y + Z}$$

$$z = \frac{Z}{X + Y + Z} = 1 - x - y$$

Chromaticity diagram: project onto 2D



Commonly-used color spaces

CIE $L^*a^*b^*$: perceptually uniform, changes in the values correspond more closely to perceived differences in color. Scaling of axes to represent "color distance"

RGB: digital displays and imaging

CMYK: printers; **HSL**, **HSV**, **HSK**, etc...

conversion between color spaces:

<https://medium.com/@alakhsharmacs/rgb-to-lab-color-space-conversion-formulas-insights-and-applications-7fd6efa519dd>

